

classes is very consistent. In almost every period, T-bills had the lowest volatility in every period, followed by corporate bonds, then large stocks, and finally small stocks.

As mentioned earlier, price volatility alone is not a reason to expect high returns. Just because something goes up and down in price doesn't make it a good long-term investment. Price changes can occur for many reasons, such as changes in the expectation of earnings and dividend growth for a company, the changing expectation of future inflation for a bond, and the expected supply and demand for commodities.

In the financial world, risk is the primary driver of return. The average standard deviation gives investors a rough idea of the relationship between the historic risk and expected return of one asset class relative to the historic risk and return of another. It should be no surprise that in comparing stocks, bonds, and T-bills, that stocks have had higher long-term return because of the higher risk. [Table 2-1](#) illustrates this phenomenon.

[Table 2-1](#) has three return columns. The simple average return column on the left is computed by summing the returns of all years and dividing by 60. The compounded return to the right is derived by linking the return in one year to that in the next in a continuous chain. The compounded return is also known as the annualized return. The last column on the right is the difference between the simple average return and the compounded return. This is the impact of price volatility on compounded returns.

**TABLE 2-1**

Asset-Class Data from 1950 to 2009

| Asset Class                     | Simple Average Return | Standard Deviation ( $\sigma$ ) | Compound Return | Volatility Return Loss |
|---------------------------------|-----------------------|---------------------------------|-----------------|------------------------|
| Treasury bills                  | 4.8%                  | 0.8%                            | 4.7%            | -0.1%                  |
| Long corporate bonds            | 6.6%                  | 8.6%                            | 6.2%            | -0.4%                  |
| Large U.S. stocks (decile 1-2)  | 12.0%                 | 18.1%                           | 9.3%            | -2.7%                  |
| Small U.S. stocks (decile 6-10) | 15.3%                 | 27.5%                           | 11.2%           | -4.1%                  |

Source: CRSP Database, Federal Reserve